Ming Huang

(626)342-7532|Email|Portfolio|LinkedIn

EDUCATION

Cornell University, *Ithaca, NY* *Aug 2025 - Present*

- M.Sc. Mechanical Engineering, *minor in Robotics*

Michigan State University, *East Lansing, MI* *Sep 2021 - May 2025*

- B.S. Mechanical Engineering, *concentration in Biomedical Engineering (GPA: 3.7/4.0, Dean's List)*

RESEARCH & ENGINEERING EXPERIENCE

Collective Embodied Intelligence Laboratory, Department of Electrical and computer Engineering, Cornell University

Robotics R&D – Advisor: Dr. Kirstin H. Petersen *Aug 2025 - Present*

- Building a fully autonomous aquatic robot for invasive vegetation management.
- Currently integrating an underwater manipulator end effector for autonomous detection, cutting, and collection of submerged vegetation. **[Details]**

Rev: Ithaca Startup Works

Hardware Engineering & Prototyping instructor *May 2026 - Aug 2026*

- Supported 16 Startups by developing concept into functional prototypes across MedTech, AgTech and GeneralTech.
- Designed, built, and integrated electromechanical prototypes across mechanical mechanisms, embedded systems, protoboard electronics, fabrication, and iterative testing. **[Details]**

Prism Controls (PMSI)

Mechanical Engineering intern *May 2024 - Aug 2024*

- Independently developed BarnFlowDynamics, a large scale CFD framework for airflow and heat transfer analysis in agricultural facilities using OpenFOAM.
- Led end to end simulation workflows on HPC, including meshing, turbulence and thermal transport modeling, solver configuration, parallelization, and convergence optimization. **[Details]**

Neural Engineering Laboratory, Michigan State University

Research Assistant – Advisor: Dr. Galit Pelled *Aug 2024 - Oct 2025*

- Investigated neural circuitry and motor control in octopus arms through stimulation experiments using high density multi electrode arrays.
- Developed analysis pipelines for signal processing, spike detection, and neural data visualization. **[Poster]**

Computational Biomechanics Laboratory, Michigan State University

Research Assistant – Advisor: Dr. Lik Chuan Lee *Aug 2024 - May 2025*

- Developed finite element models of swine hearts to study ventricular aortic interactions and cardiac mechanics.
- Built and refined 3D meshes from echo data, defined myocardial fiber architecture, and prepared datasets for electromechanics simulations. **[Poster]**

PUBLICATIONS

1. **Huang, M.,** Wang, J., Yu, L., Feng, R., He, Y. (2021). *Using speech recognition to analyze the physical condition of the elderly and its application in telemedicine. International Core Journal of Engineering*, 7(12), 619–626. DOI:10.6919/ICJE.202112_7(12).0086

TEACHING EXPERIENCE

Teaching Assistant – MAE2250: Intro to Mechanical Design, Cornell University *Jan 2026 - May 2026*

- Operate lab sections and helping students with CAD, prototyping, and mechanical design project.
- Guide students through the design process, troubleshooting fabrication issues and manufacturing planning.

TECHNICAL SKILLS

- **Mechanical Design & Fabrication:** Fusion 360, SolidWorks, NX, mechanism design, machining, waterjet, laser cutting, FDM/SLA 3D printing.
- **Electronics & Embedded Systems:** MCU, SBC, protoboard electronics, sensor/actuator integration, LabVIEW, PSpics.
- **Robotics & Programming:** Python, C++, MATLAB, ROS, Git.
- **Simulation & Modeling:** CFD, FEA.